

Answer **all** the questions.

- 1 (a) (i) Explain why the boiling tube is left in the water-bath for at least 3 minutes in step 3.

- So that M and C will reach/ equilibrate to the temperature of water bath/ 35°C to 40°C (which is the **optimum temperature** for coagulation/ the reaction/ hydrolysis of proteins in milk)

*Reject: equilibrate to same temperature without mention of water bath or water bath temperature*

[1]

- 9 Record in (a)(ii) the time taken to reach the end-point.

If the end-point has not been reached in 4 minutes, stop the experiment and record 'more than 240'.

- (ii)
- Time taken in a suitable range less than 240s / ignore d.p of time
  - units in s/ sec/ seconds

Time taken to reach end-point..... [1]

- (ii) A significant source of error for this investigation is deciding when the end-point is reached.

Suggest **one** advantage of carrying out this trial test.

[1]

- Idea of : (learning to identify when the end-point is reached) so that the same standard (of end point) can be applied to subsequent/ future/ actual experiments / reference point/ identify end-point consistently

*Reject for comparison*

- (b) (i) Complete Table 1.1 to show how you will make the concentrations of the calcium chloride solutions, **C1**, **C2**, **C3** and **C4**, and how you will set up the control, **T**.

**Table 1.1**

|                                                                                    | <b>C</b> | <b>C1</b> | <b>C2</b> | <b>C3</b> | <b>C4</b> |
|------------------------------------------------------------------------------------|----------|-----------|-----------|-----------|-----------|
| Concentration of calcium chloride /%                                               | 20.0     | 10.0      | 5.00      | 2.50      | 1.25      |
| Label of calcium chloride to be diluted                                            |          | C         | C1        | C2        | C3        |
| Volume of the calcium chloride solution to be diluted / cm <sup>3</sup>            |          | 5.0       | 5.0       | 5.0       | 5.0       |
| Volume of the distilled water, <b>W</b> , to make up the dilution/ cm <sup>3</sup> |          | 5.0       | 5.0       | 5.0       | 5.0       |
| Description of the control, <b>T</b> :                                             |          |           |           |           |           |

- T1: selects concentrations at 10.0, 5.00, 2.50 and 1.25; *All concentrations must be given to three significant figures.*
- T2: for each dilution, adds same volume of water as volume of correctly selected calcium chloride solution; Values should be recorded consistently and to the appropriate precision (1dp)
- T3: makes up each solution to have suitable volume to use i.e. 10 cm<sup>3</sup> or more (after all solutions prepared);
- T4: Control: replace calcium chloride with **same volume** of distilled water, subjected to same experimental condition

Reject: boiled enzyme as the control is for calcium chloride.

[4]

**17** Repeat step **16** with each of the other boiling tubes.

(ii) Record your results in an appropriate manner in the space below.

| Concentration of calcium chloride solution / % | Time taken to reach end-point / s |
|------------------------------------------------|-----------------------------------|
| 20.0                                           | shortest                          |
| 10.0                                           |                                   |
| 5.00                                           |                                   |
| 2.50                                           |                                   |
| 1.25                                           | Longest                           |
| 0.00 (T)                                       | More than 240                     |

- R1: Complete lined table drawn with suitable & complete column/row headings with correct units (Concentration of calcium chloride solution / % & Time taken to reach end-point / s); Reject: sec/ seconds as proper units
- R2: Time taken recorded as seconds, to whole number (ignore d.p. for concentration of calcium chloride since it has been marked in (b)(i))
- R3: records shortest time for highest concentration of calcium chloride solution to longest time for lowest concentration;
- R4: records control as 'more than 240'; Reject: 4mins

[4]

(ii) Calculate the rate of coagulation of milk when the concentration of calcium chloride was 20.0%.

- 1 mark for correct calculation of  $1 / \text{time}$  with unit  $\text{s}^{-1}$ ;
- Answers should be given to an appropriate number of significant figures depending on the sf of the time taken to reach end-point for 20% calcium chloride

Rate of coagulation..... [1]

(iv) Suggest an explanation of the effect of calcium ions on the coagulation of milk by enzyme E.

- calcium ions **change shape of active site** (*reject if did not mention active site*) so proteins can fit more easily/ more complementary in shape to substrate / AVP: mentions some effect on the enzyme structure  
(*note: the substrate molecules here are proteins - not "milk molecules", amino acids*)  
*Reject merely saying higher affinity or merely saying stabilising the active form of the enzyme w/o saying HOW*

AVP: calcium ions is a **cofactor** of enzyme E

AVP: proposed mechanisms on how cofactors work e.g. calcium ions bind and help to lower activation energy of reaction/ change the charges on the contact and/or catalytic residues

- more calcium ions means more **enzyme-substrate complexes** can form (per unit time), so milk clots faster

*Reject: Non-comp inhibitor/ comp inhibitor → makes active site more complementary to shape (contradiction!)*

[2]

(v) Explain why temperature is a significant source of error in this investigation and suggest a modification to reduce or eliminate this error.

- temperature affects the **kinetic energy** of enzymes and substrates , affecting **effective collisions** between E and S  
*Reject merely saying temperature will affect enzyme activity / enzyme are temperature sensitive without using biological knowledge to explain why*
- Conduct the experiment using a **thermostatically-regulated waterbath** to ensure **constant temperature** throughout the experiment  
( idea of maintaining constant temperature / preventing fluctuating temperature must be present)

Accept:

- Since the enzyme and substrate are not at the same temperature before adding them together, so when they are added together, the medium temperature become a factor in affecting rate of enzymatic reaction/ the results are not solely due to the amount of  $\text{CaCl}_2$ , could be due to temperature differences between E and S
- Equilibration** of the enzyme in the waterbath also to ensure that temperature of the enzyme and substrate solutions is the **same** before adding them together;

*Reject insignificant points: While 1 tube is taken out of the waterbath to carry on the reaction, the other tubes still remaining inside will reach a higher/ lower temperature / 3 minutes incubation time is too short*

[2]

- (c) (i) This procedure can be modified to investigate other independent variables.

Describe how you could modify this procedure to investigate the effect of pH on the time taken to reach the end-point.

- Use a standardised/ fixed calcium chloride concentration  
*Reject replace the varying concentrations of calcium chloride with HCl/NaOH- not clear if CaCl<sub>2</sub> is still added to the boiling tubes. Seems to suggest that calcium chloride is not added into the boiling tubes at all.*
- ref. to use of different **buffer** to achieve and maintain different pH in different tubes  
*Reject: vary pH by varying amount of HCl / NaOH / acid / alkaline ( since pH is also a significant factor that affects E activity, pH must be maintained throughout the experiment by use of a buffer)*

(keeping all other experimental conditions the same)

[2]

- (ii) Fig. 1.4 shows the results of the investigation done by a student.

|      |        |        |        |        |      |
|------|--------|--------|--------|--------|------|
| pH 2 | 215 s  | pH 3.5 | 15 s   | pH 4.9 | 75 s |
|      | pH 2.7 | 60 s   | pH 6.2 | 220 s  |      |

**Fig. 1.4**

Plot, on the grid, a graph of the student's results in Fig. 1.4 to show the effect of pH on the time taken for milk to coagulate. Draw a line of best fit.

[3]

- G1: correct **orientation** of axes, complete with **labels and units**, and **intervals** along axes correctly **labelled**: independent variable – pH on x-axis, dependent variable time taken for milk to coagulate/ s on y-axis
- G2: **sensible scale** in both the x and y directions and all points **plotted accurately** within half a small square;  
*If the scale is awkward the points do not need checking.*
- G3: a smooth curve that passes through all the plotted points - no extrapolation

- (iii) Estimate the time taken to reach the end-point at pH 5.3.

- correct reading from the graph with units;  
*Reading should be accurate to within half a small square.*

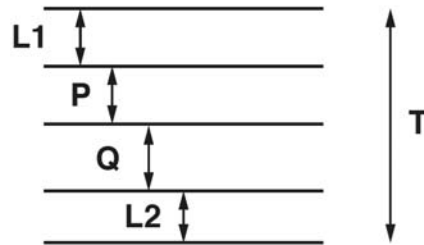
[1]

[Total: 22]

- 2 (a) Select a part of the leaf on **N1** which shows the four layers **L** (**L1** and **L2**), **P** and **Q**.

Do **not** include a vascular bundle.

- (i) Use the eyepiece graticule in the microscope to measure:
- the depth of the whole leaf, **T**
  - the depth of each of the tissues, **L** (**L1** and **L2**), **P** and **Q**, as shown in Fig. 2.1.



**Fig. 2.1** (not drawn to scale)

**T** = ..... eyepiece graticule units

**L1** = ..... eyepiece graticule units

**P** = ..... eyepiece graticule units

**Q** = ..... eyepiece graticule units

**L2** = ..... eyepiece graticule units

[3]

- 1 states all 5 measurements (**T**, **L1**, **P**, **Q** and **L2**);  
 2 **L1** and **L2** have to be smaller values than **P** and **Q**;  
 3 measurement of **T** = sum of other measurements;

- (ii) Use the measurements from (a)(i) to determine the simplest ratio of the depth of the leaf (**T**) to the depth of the palisade layer (**P**).

You may lose marks if you do not show your working.

- 1 uses measurements of epg units **T + P** or **Q** (whichever is smaller) ;  
 2 Larger number to smaller number ;  
 3 To lowest common denominator ;

simplest ratio ..... [3]

Use a sharp pencil for drawing.

- (iii) Use the measurements from (a) (i) to help you draw a large plan diagram of a part of the leaf on **N1**, as shown by the shaded area in Fig. 2.2.

This must include at least **one** vascular bundle and **one** stoma.

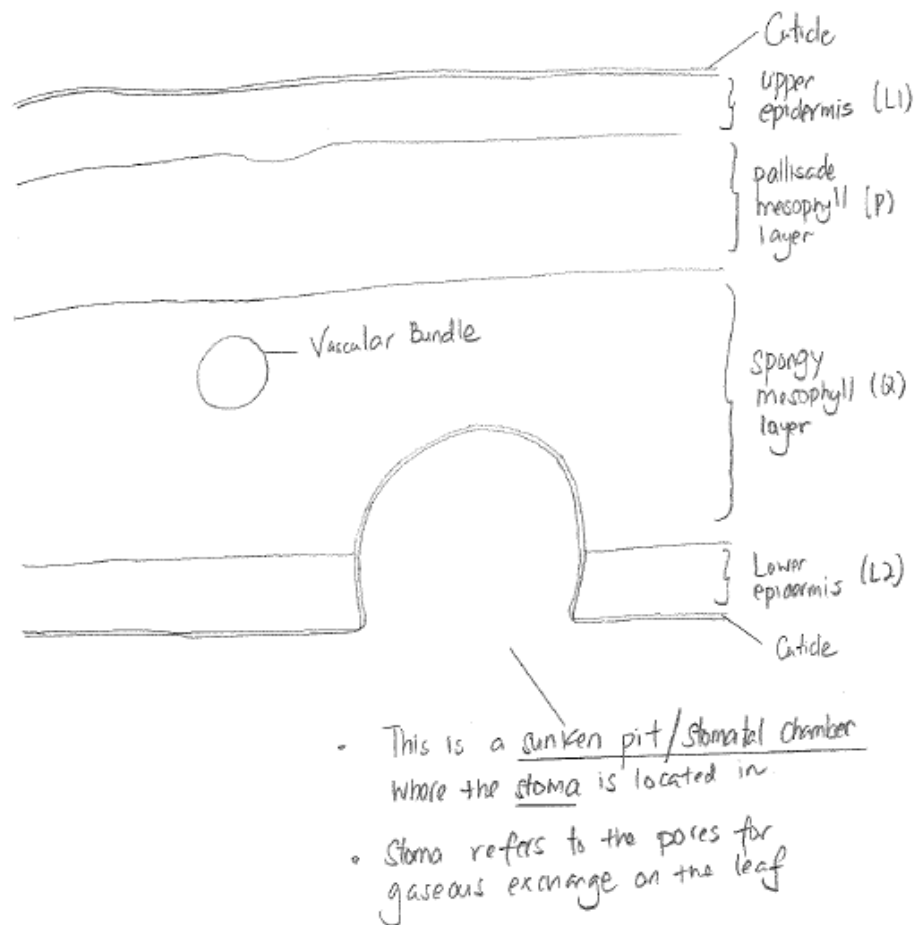
*You are expected to draw the correct shape and proportions of the different tissues.*

Use **one** ruled label line and label to identify the palisade layer.

## Marking criteria

- 1) Drawing must be larger than  $\frac{1}{2}$  the space give (at least 90mm in length x 70mm in breadth) + at least 4 different layers (according to fig 2.1) excluding the upper and lower epidermal layers + no shading;
- 2) No individual cells + at least one vascular bundle of correct shape and proportion+ one sunken pit of correct shape and proportion, where there stoma is located in;
- 3) correct proportion of palisade to whole depth of leaf, ratio is 25 – 50% of the entire depth;
- 4) thin Upper and lower layer of cuticle drawn.
- 5) uses one label line + one label "P" ;

[5]



(iv) Observe the upper epidermis of the leaf on **N1**. The upper epidermis has no guard cells.

Select one group of **four** adjacent (touching) cells from the upper epidermis.  
Each cell must touch at least one of the other cells.

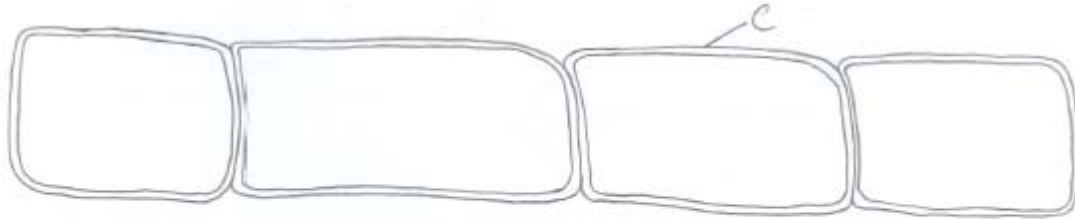
Make a large drawing of this group of **four** cells.

Use **one** ruled label line and the label **C** to identify a structure made of cellulose.

1. **Smooth, unbroken, clear lines** + minimum size at least 40 mm across the largest cell + the length of the cells should be longer than the width in at least one of the cell (correct shape) + no shading;
2. Only **four cells drawn (can be in a line or in layers)** + each cell touching at least one other cell + cells are joined together correctly.

3. cell wall drawn as two lines close together + even thickness of the cell wall for all the cells ;
4. uses one label line + one label "C";

### Sample drawing



HP Drawing of four upper epidermis cells (x400)

[4]

- (b) Fig. 2.3 is a photomicrograph of a stained transverse section through part of a plant leaf from a different type of plant.

You are not expected to be familiar with this specimen.

- (i) In Fig. 2.3 the lines **P**, **Q**, **R** and **S** are drawn across the length of four vascular bundles.

Describe how you would find out the **mean** actual length of these four vascular bundles.

- Use ruler to measure the lines **P**, **Q**, **R** and **S** are drawn across the length of four vascular bundles;
- In cm or mm;
- divides by 59 (magnification);
- add up **P**, **Q**, **R** and **S** divide by 4 (*reject simply saying "calculate the average" of the 4 lengths*)

1/2marks each

*Examiners' comments:*

*Some students gave detailed description on how eyepiece graticle and stage micrometer could be used to find out the mean length of the vascular bundles. This is incorrect because you are not given the actual slide at all to do the measurement.*

[2]

- (ii) Observe the leaf on **N1** and the leaf in Fig. 2.3.

Use a suitable table to record observable differences between the specimen in Fig. 2.3 and the specimen on slide N1.

Any 2 differences

| Feature<br><i>optional- good to have</i>                  | <b>Fig. 2.3</b><br><i>This is a monocot plant.<br/>Hence, the leaf structure<br/>is different from a dicot<br/>plant.</i> | <b>Slide N1</b><br><i>This is a dicot plant that<br/>grows in <u>dry condition</u>.<br/>Therefore, it has<br/>adaptions that allow it to<br/>survive well in dry<br/>condition</i> |
|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Size of vascular bundle relative to the depth of the leaf | Some vascular bundles occupy the entire width of leaf.                                                                    | Vascular bundles occupy a smaller width of the leaf.                                                                                                                               |
| Mesophyll layer                                           | One layer of mesophyll layer.                                                                                             | Two different layers of mesophyll cells. (palisade and spongy)                                                                                                                     |
| Mesophyll layer                                           | Cells are densely packed in a mesophyll layer.                                                                            | Cells are loosely packed in a mesophyll layer.                                                                                                                                     |
| Presence of air spaces                                    | Air space absent/few                                                                                                      | Air spaces present                                                                                                                                                                 |
| Palisade layer                                            | No palisade layer.                                                                                                        | Presence of palisade layer. (longitudinal cells)                                                                                                                                   |
| Stomata in pit                                            | Stomata are not located in a pit.                                                                                         | Stomata found in a pit                                                                                                                                                             |
| Presence of hair                                          | Hair is absent.                                                                                                           | Hair are present in the stomatal chamber/sunken pit.                                                                                                                               |
| Location of stomata                                       | Stomata (plural) present on both upper and lower epidermis.                                                               | Stomata (plural) present on the lower epidermis.                                                                                                                                   |
| Cuticle                                                   | Presence of thick cuticle on the surfaces of the leaf.                                                                    | Absence of cuticle on the surface of the leaf.                                                                                                                                     |
| Epidermal layer                                           | Upper epidermal layer is not continuous, interrupted with groups of cells.                                                | Upper epidermal layer is continuous.                                                                                                                                               |
| Epidermal layer                                           | Upper and lower epidermal layer is thin. (1 layer each)                                                                   | Upper and lower epidermal layer is thick. (3 layers)                                                                                                                               |

*Examiners comment:*

- *Answers related to “numbers” and “size” are all largely rejected. Eg: comparing the number and size of vascular bundles, number of stomata, size of airspaces.*
- *This is because no actual counting or measurement of the size are done. The differences in the numbers and sizes could actually be due to different magnification of images. (Fig 2.3 has a magnification of 59x while you could have used 100x in viewing specimen N1. Moreover, the areas viewed under the microscope are different as well.*

3

| Tube | percentage concentration of carbon dioxide | colour |
|------|--------------------------------------------|--------|
| X    | increasing above 0.04                      | orange |
| Y    | 0.04<br>(normal atmospheric)               | red    |
| Z    | falling below 0.04                         | purple |

[1]

Table 3.1



- (b) Suggest why the indicator will turn orange when the bench lamp is placed at a distance greater than 20 cm from the setup.

- *Light intensity level where respiration will be greater than photosynthesis rate;*

[1]

*Some students mentioned that the plant would not photosynthesise but failed to fully explain why CO<sub>2</sub> concentrations would increase.*

Using the information given and your own knowledge, design an experiment to investigate the effect of temperature on the rate of photosynthesis of pondweed (*Cabomba* sp.)

[12]

**Theory of explanation [0.5m each, max 1m]**

- T1     Photosynthesis / carbon fixation is an Rubisco-controlled / enzyme-controlled reaction;
- T2     The higher the temperature, increase in **kinetic energy** of enzyme and substrate leads to increased more CO<sub>2</sub> binding to Rubisco / **enzyme-substrate complex formation**;
- T3     More CO<sub>2</sub> fixed per unit time / higher rate of carbon fixation, faster the CO<sub>2</sub> concentration in the solution decreases to change the colour of indicator;

Accept inverse relationship if theory of enzyme **denaturation** at higher temperature (beyond optimum) is mentioned

**Hypothesis [0.5m each, max 1m]**

- H1     The higher the temperature, the higher the rate of photosynthesis in the pondweed;
- H2     The faster / the shorter the time taken for the indicator to turn purple in colour;
- or
- the shorter the time taken for the solution to reach an absorbance value of X (measured using the colorimeter);
- or
- the higher the absorbance value of the solution at the end of 5 minutes of light exposure (measured using the colorimeter);

Accept inverse relationship if theory of explanation is regarding denaturation, with specific reference to temperature beyond optimum.

**Independent, Dependent, Controlled Variables [0.5m each, max 2m]**

- V1     **Independent variable:** temperature (28°C, 29°C, 30°C, 31°C, 32°C)  
Data points should be equally spread out and ideal range is 28°C to 32°C.
- V2     **Dependent variable:** time taken for indicator solution to turn purple, measured by the stopwatch;
- V3     **Controlled variable:** Light intensity
- V4     **Method to keep light intensity constant:** Fixing the distance at which the lamp is placed from the setup by measuring with a ruler and using the same lamp throughout the entire experiment;
- V6     **Rationale** for distance of the lamp from the setup as a controlled variable: ref. to the idea that Light intensity will be varied by changing the distance at which the

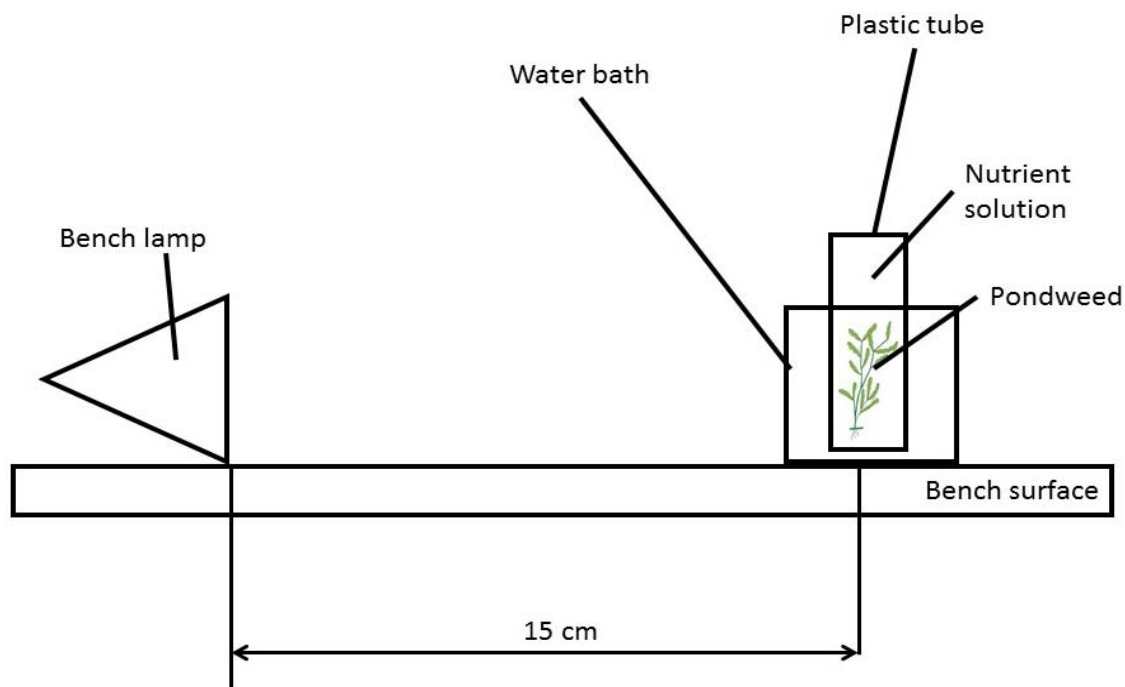
lamp is placed from the setup so using the same lamp and fixing distance will ensure light intensity stay constant and will not affect rate of photosynthesis;

- V7 **Controlled variable:** no. of leaves / plant used for measuring photosynthesis rate;
- V8 **Method to keep no. of leaves / plant used for measuring photosynthesis rate constant:** using the same pondweed plant for all experiments;
- V9 **Rationale** for keeping no. of leaves / plant used for measuring photosynthesis rate constant: ref. to the idea that different number of leaves will lead to different number of chloroplasts which are photosynthesizing, affecting the rate at which CO<sub>2</sub> concentration decreasing in the solution;
- V10 Any other valid variable;
- V11 Any other valid variable;

**Method [0.5m each, max 3m] + Diagram [0.5m each, max 1m] - Replaces some steps written in text**

- M1 Using 10 cm<sup>3</sup> syringe, measure 8cm<sup>3</sup> (*accept any volume lower than 10 cm<sup>3</sup>*) of the nutrient solution into a plastic tube;
- M2 Place one pondweed plant into the tube;
- M3 Prepare a beaker as the water bath by mixing boiling water and cold water to adjust the water temperature to 28°C / Constantly remove water from the water bath beaker and add boiling water into the water bath to maintain the temperature of the water bath at 28°C;
- M4 Use the **thermometer** to measure the temperature of the water bath to ensure that it is 28°C;
- M6 Place the plastic tube containing the plant and solution into the water bath and using the **stopwatch**, time for **5 minutes**;
- M7 This is for **equilibration** to ensure that the plant is at 28°C before experiment begins;
- M8 Using the ruler, measure 10 cm (*accept any reasonable distance no more than 20 cm*) away from the water bath and place the bench lamp;

- M9 Using a 5 cm<sup>3</sup> syringe, add 2 cm<sup>3</sup> of indicator solution into the plastic tube and sealed it with the lid.
- M10 Switch on the lamp and using the **stopwatch**, record the **time taken for the indicator solution in the plastic tube to turn purple**
- or
- time taken for indicator solution to reach the final absorbance value of purple indicator solution**
- or
- measure the absorbance value of the indicator solution at the end of 5 minutes (accept reasonable amount of time);**
- M11 Measure the **initial absorbance** of the indicator of the solution using a colorimeter / spectrophotometer;
- M12 **Method mark** for measurement using spectrophotometer (description of the method must make sense in its entirety)
- Every 1 minute (accept reasonable amount of time), use a 1 cm<sup>3</sup> syringe to draw out 0.5 cm<sup>3</sup> of solution (accept reasonable amount of solution, bearing in mind the tube only can have maximum of 10 cm<sup>3</sup> of solution) and measure the absorbance of the indicator using the spectrophotometer / colorimeter;
- or
- At the end of 5 minutes, use 1 cm<sup>3</sup> syringe to transfer 1 cm<sup>3</sup> of indicator solution into cuvette and measure the absorbance of the solution using spectrophotometer / colorimeter;
- M13 Repeat the experiment at other temperatures (28°C, 29°C, 30°C, 31°C, 32°C);
- M 14 Conduct **three replicates** at each temperature to obtain an **average time taken for the indicator solution to turn purple** and **repeat the entire experiment twice** to ensure **reproducibility**;
- D1 Diagram of correct setup;
- D2 Well labelled;



**Control [0.5m each, max 1m]**

- C1 Description: Repeat the experiment but replace the pondweed plant with a boiled pondweed plant which has been immersed in boiling water for 5 minutes;
- C2 Result: The indicator solution remains red in colour;  
Reject: no colour change in the indicator
- C3 Rationale: This is to show that the change in colour of indicator solution is due to the pondweed plant (photosynthesising) and not any other factors;

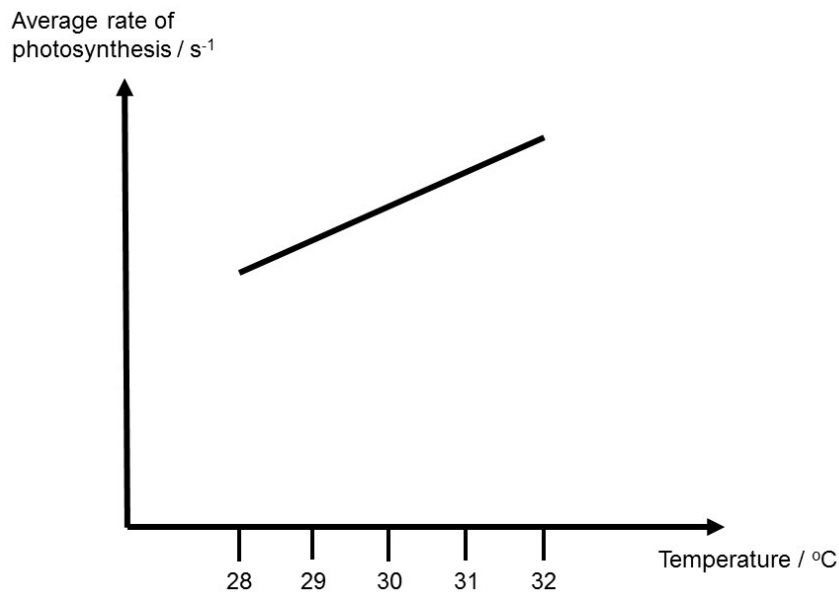
**Results [0.5m each, max 2m]**

- R1 suitable column / row headings with units (time taken for indicator solution to turn purple / s, temperature / °C, average rate of photosynthesis / s<sup>-1</sup>);
- R2 Includes replicates, average and processing of data (calculation of rate of photosynthesis);

| Temperature / °C | Time taken for indication solution to turn purple / s |                |                |                  | Rate of photosynthesis / s <sup>-1</sup> |
|------------------|-------------------------------------------------------|----------------|----------------|------------------|------------------------------------------|
|                  | T <sub>1</sub>                                        | T <sub>2</sub> | T <sub>3</sub> | T <sub>avg</sub> |                                          |
| 28.0             |                                                       |                |                |                  |                                          |
| 28.8             |                                                       |                |                |                  |                                          |
| 29.6             |                                                       |                |                |                  |                                          |
| 30.4             |                                                       |                |                |                  |                                          |
| 31.2             |                                                       |                |                |                  |                                          |

- R3 Graph, with the axis correctly labeled (y axis: average rate of photosynthesis / s<sup>-1</sup>, x axis: temperature / °C);

- R4 Trend showing higher rate of photosynthesis at higher temperature (Accept if trend showed optimum temperature providing that students used scale to show no extrapolation and also optimum)



**Statistical method [0.5m each, max 1m]**

- S1 Conduct (two-tailed one sample) T test for average rate of photosynthesis at 28 $^{\circ}\text{C}$  and 32 $^{\circ}\text{C}$ ;
- S2 If the If critical  $-t$  value  $<$  test statistic  $<$  critical  $t$  value at  $p = 0.05$  (two-tailed), do not reject  $H_0$  that there is no significant difference between average rate of photosynthesis at 28 $^{\circ}\text{C}$  and 32 $^{\circ}\text{C}$ ;

**Safety / Risk Assessment [0.5m each, max 1m]**

- S1 Risk of electrical shock when handling the beach lamp;
- S2 Ensure hands are dry when handling the beach lamp;
- OR
- S3 Indicator solution is an irritant to eyes and skin;
- S4 Wear safety goggles and gloves to avoid contact with eyes and skin;